

Can a Mechanistic Digital Twin Be Both Reliable and Controllable?

A frozen-evidence synthesis of M1 and C1

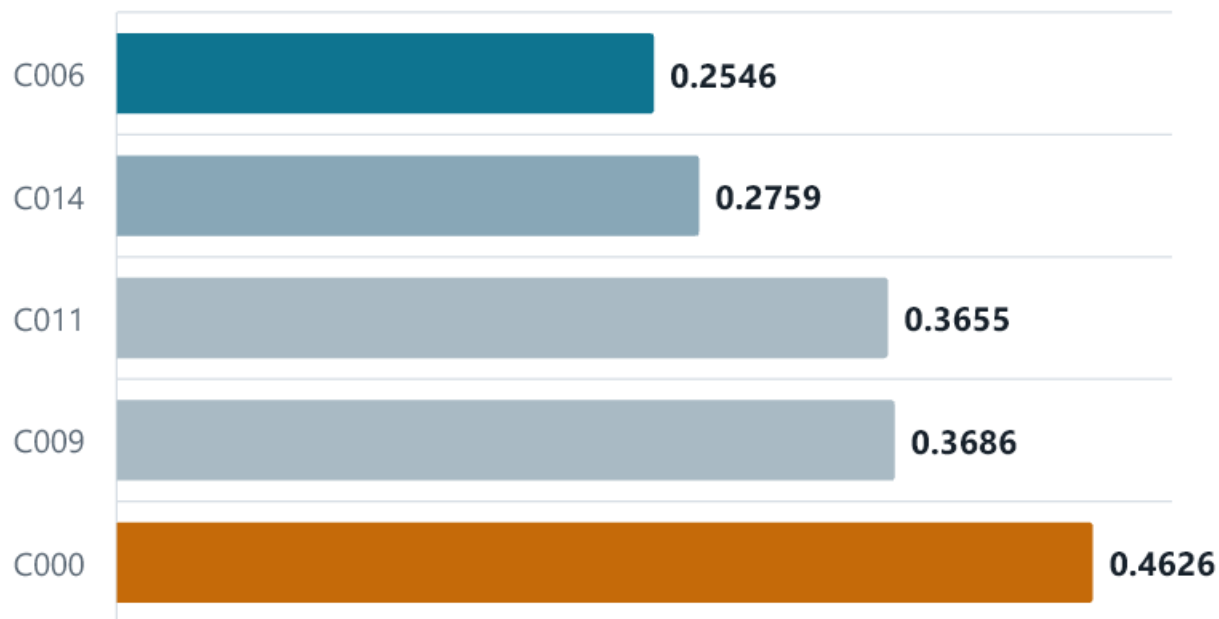


M1 explains why C006 is the frozen twin. C1 tests whether that same C006 responds reproducibly to bounded causal control.

C006 emerged under the frozen M1 selection rule

FIT selection

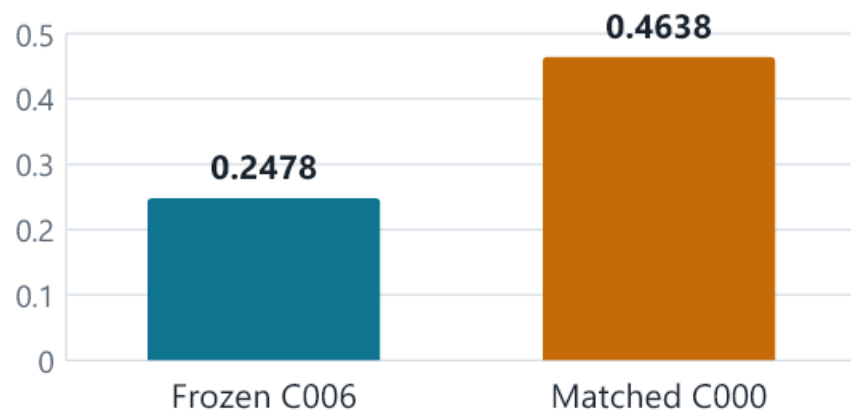
Lowest NRMSE after FIT-only affine calibration



All 16 candidates were evaluated; the four lowest plus matched C000 are shown.

Frozen HELDOUT check

C006 was not reselected after HELDOUT access

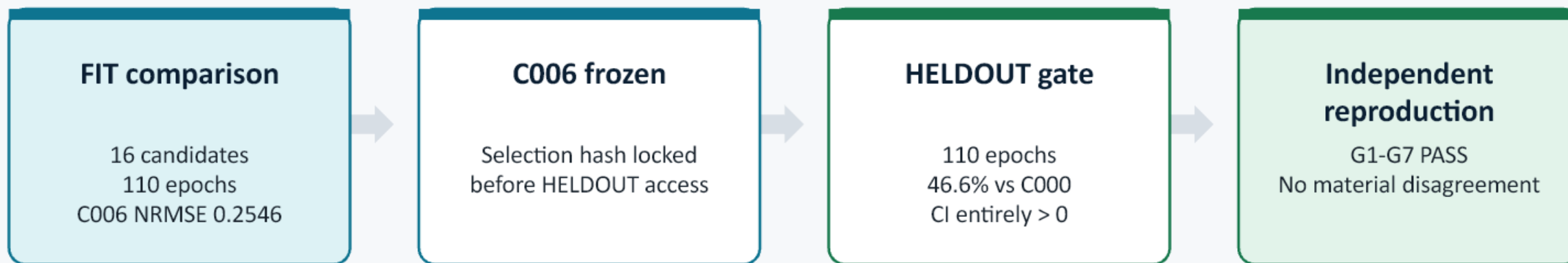


46.6%

relative HELDOUT NRMSE improvement

95% paired bootstrap CI: [0.0798, 0.1015]

The reliable-twin chain was frozen before controller research



Provenance carried into C1

protocol

338366...E3C3

model

775A39...3869

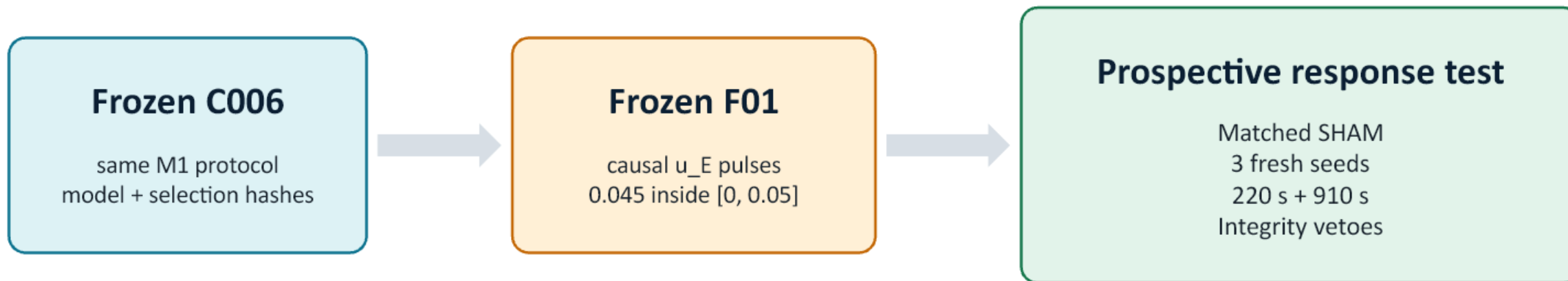
selection

3D8C3D...8BBF

Conclusion

C006 was frozen before controller research.

A reproducible T2 response defines controllability



Target hierarchy

PRIMARY

T2

paired relative response
desired direction > 0

SUPPORTING

I_SO

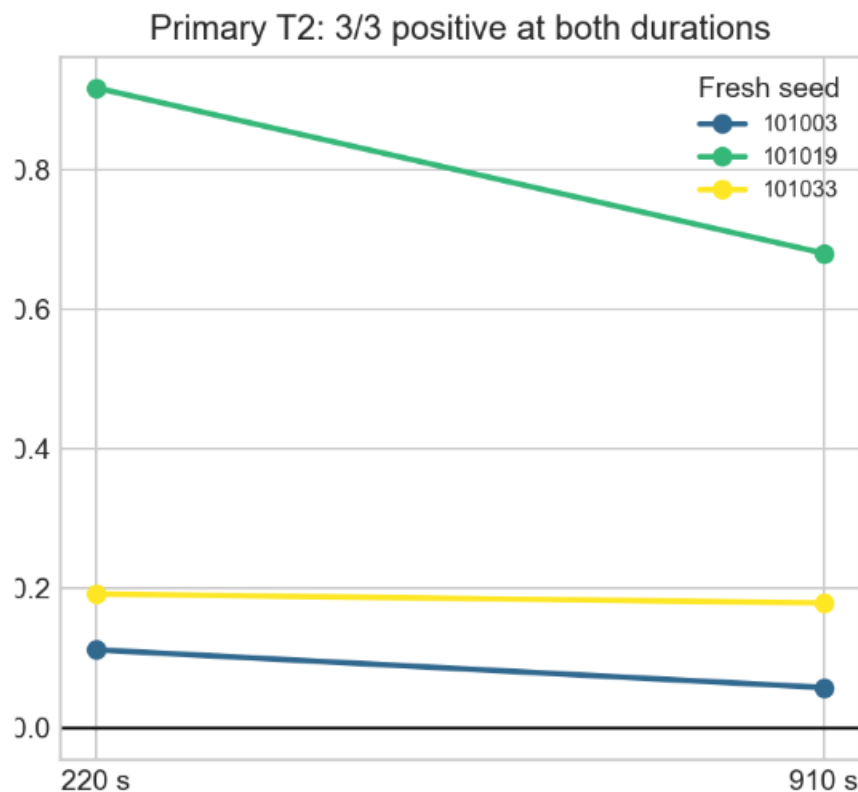
independent corroboration

SECONDARY - VISIBLE, NOT PRIMARY

N_SO_MIN | I_SP | N_SP_MIN
P_SO | T6 | P(C|SP)

Adverse secondary effects remain in the claim boundary.

Primary T2 response replicates across seeds and duration



Authoritative C1 quicklook crop - paired dT2 (%)

220 s

+0.1913%

median dT2

3/3 positive

910 s

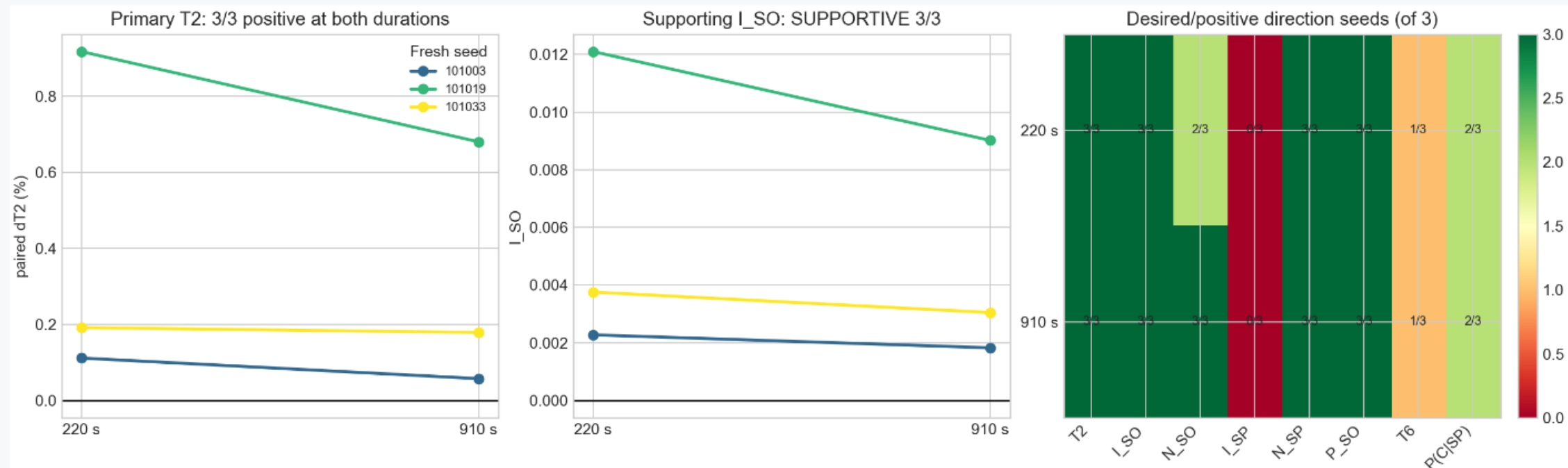
+0.1784%

median dT2

3/3 positive

Integrity PASS 6/6

I_SO corroborates T2; secondary trade-offs remain



Authoritative C1 quicklook - supporting and secondary panels

I_SO

supportive 3/3 at both durations

I_SP

adverse 3/3 at both durations

T6

mixed; slightly negative medians

Primary controllability is established; controller-quality trade-offs remain.

All required components of the bounded MVP passed

Reliable C006	M1 selection + held-out + reproduction	PASS
Frozen provenance	M1 identities retained in all six C1 runs	PASS
Frozen causal F01	No retuning during fresh confirmation	PASS
Bounded legal actuator	$u_E = 0.045$ inside $[0, 0.05]$	PASS
Fresh multi-seed T2	3/3 positive at 220 s	PASS
910 s duration robustness	3/3 positive at 910 s	PASS
I_SO corroboration	3/3 supportive at both durations	PASS

Final bounded claim**Reliable + Controllable Digital Twin MVP**

Advisor decision: close Part II at the bounded MVP level?

Evidence supporting closure

- C006 selected by a frozen rule and held-out advantage
- Protocol, model, and selection identities retained in C1
- T2 positive 3/3 across fresh seeds at 220 s and 910 s
- Causal bounded F01 and integrity PASS 6/6
- I_SO independently supportive at both durations

Limitations that remain

- I_SP adverse in every fresh run
- T6 mixed with slightly negative medians
- One-subject Night-1, proxy-level M1 scope
- No clinical, population, global, or all-endpoint claim
- Future work: spindle/coupling-aware controller quality

**Close Part II at the bounded in-silico Reliable + Controllable
Digital Twin MVP level?**